Bonosri Ghose
Google Scholar | LinkedIn

✉ bonosri.bgd@gmail.com/
bonosri.ghose@student.uclouvain.be

WORK EXPERIENCE

- **Intern,** VITO (Flemish Institute for Technological Research), Boeretang 200, 2400 Mol, Belgium (1st July 2025 – 31st August 2025)
  Title of the Internship: Evaluating the European Agricultural Area
  Responsibilities
  - Evaluating and validating the High-Resolution Layers (HRL) Vegetated Land Cover Characteristics (VLCC) products using both visual interpretation (e.g., consistency) and statistical methods (e.g., IQR analysis).
  - Preparing available reference data for defined areas with lower performance, with an emphasis on interpreting high-resolution imagery for data collection.

- **Research Assistant,** Disaster Management E–Learning Centre, Begum Rokeya University, Rangpur, Bangladesh (1 May 2019 – 1 May 2021).

  Responsibilities

  - Data collection (FGD, KII, IDI), Data Analysis, and Data Interpretation.
  - Method Selection for a Particular problem
  - Literature Reviewing
  - Result writing

EDUCATION

September 2025 – to present	**Master of Science** (2nd year) in Erasmus+ program in Geo-Information Science and Earth Observation for Environmental Modelling and Management (GEM) for Food Security, Université catholique de Louvain, Belgium. **Thesis Title:** Crop type mapping using Sentinel-1 and Sentinel-2 in the smallholder cropping system of the cotton belt in Mali. **Selected Coursework:** • Spatial Modelling of Land Dynamics • Land Monitoring by Advanced Satellite Remote Sensing • Applied Geomatics • Elements of Agroecology • Agricultural and Rural Policies • Smart Technologies for Environmental Engineering • Decision Tools and Project Management
September 2024 – June 2025	**Master of Science** (1st year) in Erasmus+ program in Geo-Information Science and Earth Observation for Environmental Modelling and Management (GEM) for Food Security, University of Lund, Sweden. **Selected Coursework and Grades:** • Geographical Information Systems – Basic Course (15 ECTS): Pass (G) • Geographical Information Systems – Advanced Course (15 ECTS): Pass (G) • Ecosystem Modeling (15 ECTS): Pass (G) • Satellite Remote Sensing (15 ECTS): Pass with Distinction (VG)
March 2019 – October2020	**Master of Science,** Disaster Management, Begum Rokeya University, Rangpur. **Thesis Title:** Differential responses of rice yield fluctuation to climate changes in different sub-regions of Bangladesh.
April 2014 – February2019	**Bachelor of Science,** Disaster Management, Begum Rokeya University, Rangpur. **Undergraduate Thesis Title:** Spatiotemporal variation on wheat yield under climate change condition in northwestern Bangladesh.

AWARDS

- Erasmus+ scholarship 2024-2026 for the MSc degree program in Geo-Information Science and Earth Observation for Environmental Modelling and Management (GEM) for Food Security.
- Bangabandhu Merit Scholarship, 2020 (The Bangabandhu Merit Scholarship is provided to meritorious students who have achieved 1st, 2nd, and 3rd places in every session at undergraduate levels in all departments of the university).

PUBLICATIONS

As a first author

1. **Ghose, B.,** Islam, A.R.M.T., Islam, H.M.T. *et al.* Rain-Fed Rice Yield Fluctuation to Climatic Anomalies in Bangladesh. *Int. J. Plant Prod.* (2021). https://doi.org/10.1007/s42106-021-00131-x
2. **Ghose, B**., Islam, A.R.M.T., Kamruzzaman, M. *et al.* Climate-induced rice yield anomalies linked to large-scale atmospheric circulation in Bangladesh using multi-statistical modeling. *Theor Appl Climatol* (2021). https://doi.org/10.1007/s00704-021-03584-2
3. **Ghose, B**., Islam, A.R.M.T., Salam, R., Shahid, S., Kamruzzaman, M., Das, S., Elbeltagi, A. and Salam, M.A., 2021. Rice Yield Responses in Bangladesh to Large-scale Atmospheric Oscillation Using Multifactorial Model.

As a co–author

1. Salam, R., **Ghose, B**., Shill, B.K., Islam, M.A., Islam, A.R.M.T., Sattar, M.A., Alam, G.M. and Ahmed, B., 2021. Perceived and actual risks of drought: household and expert views from the lower Teesta River Basin of northern Bangladesh. *Natural Hazards*, pp.1-19
2. Talukdar, S., **Ghose, B.,** Salam, R., Mahato, S., Pham, Q.B., Linh, N.T.T., Costache, R. and Avand, M., 2020. Flood susceptibility modeling in Teesta River basin, Bangladesh using novel ensembles of bagging algorithms. *Stochastic Environmental Research and Risk Assessment*, pp.1-24
3. Shahfahad, Talukdar, S., **Ghose, B.,** Islam, A.R.M.T., Hasanuzzaman, M., Ahmed, I.A., Praveen, B., Asif, Paarcha, A., Rahman, A. and Gagnon, A.S., 2023. Predicting long term regional drought pattern in Northeast India using advanced statistical technique and wavelet-machine learning approach. *Modeling Earth Systems and Environment*, pp.1-22. DOI: 10.1007/s40808-023-01818-y
4. Mallick, J., Islam, A.R.M.T., **Ghose, B**., Islam, H.M.T., Rana, Y., Hu, Z., Bhat, S.A., 2021. Spatiotemporal trends of temperature extremes in Bangladesh under changing climate using multi-statistical techniques. Theoritical and applied Climatology.
5. Islam,A.R.M.T., Saha, A., **Ghose, B.,** Chowdhuri, I., Mallick,J., 2021. Landslide susceptibility modeling in a complex mountainous region: Application of new data mining approach. Natural Hazard
6. Islam, A.R.M.T., Al Awadh, M., Mallick, J., Pal, S.C., Chakraborty, R., Fattah, M.A., **Ghose, B**., Kakoli, M.K.A., Islam, M.A., Naqvi, H.R. and Bilal, M., 2023. Estimating ground-level PM2. 5 using subset regression model and machine learning algorithms in Asian megacity, Dhaka, Bangladesh. *Air Quality, Atmosphere & Health*, pp.1-23.
7. Mallick, J., Naikoo, M.W., Talukdar, S., Ahmed, I.A., Rahman, A., Islam, A.R.M.T., Pal, S., **Ghose, B**. and Shashtri, S., 2021. Developing groundwater potentiality models by coupling ensemble machine learning algorithms and statistical techniques for sustainable groundwater management. *Geocarto International*, pp.1-27.

RESEARCH INTERESTS

- Cropland Productivity
- Crop Type Mapping and Yield Estimation
- Climate Change Impacts in Agriculture
- Remote sensing in Agriculture
- Food Security
- Machine Learning and Geo-Spatial Analysis
- Big Earth Observational Data
- Geographic Information system (GIS)

SOFTWARE SKILLS

- Python
- Google Earth Engine
- Arc GIS pro
- Terrset

LINGUISTIC PROFICIENCY

- Mother tongue - Bengali.
- Another language - English (Professional proficiency).

ACCOMPLISHED CONFERENCE, WORKSHOP, AND WEBINAR

- The joint event with the Erasmus + MSc in Geo-Information Science and Earth Observation for Environmental Modelling and Management (GEM), from 16 June 2025 until 20 June 2025, University of Twenty, Enschede, Netherlands.
- The International Conference on Earth and Environmental Science & Technology for Sustainable Development (ICEEST) 2020 organized by the faculty of Earth and Environmental Sciences, University of Dhaka, on 25-27 January 2020, Dhaka, Bangladesh.
- A workshop on developing the draft watershed management policy and implementation framework was organized by the International Union for Conservation of Nature (IUCN) on 27 February 2020, Begum Rokeya University Rangpur, Bangladesh.

- One-day international webinar on environment, development, and sustainability, on 20th September 2020 Department of Geography, Krishna Chandra College, India.

EXTRA – CURRICULAR ACTIVITIES

- Volunteer in a brand audit of single-use plastic and observation of World Cleanup Day under the organization of Environment and Social Development Organization (ESDO) on 25 September 2019, Begum Rokeya University, Rangpur, Bangladesh.

REFERENCES

Prof. Pierre Defourny
Department of Earth and Life Institute
Universite catholique de Louvain
Louvain la nueve, Belgium
Phone: 32-10-472374.
pierre.defourny@uclouvain.be

Kasper Bonte
R&D Professional at Vito
Remote Sensing Unit
Boeretang 200, 2400 Mol, Belgium
kasper.bonte@vito.be